



A survey on Relation Extraction

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ABSTRACT

With the advent of the Internet, the daily production of digital text in the form of social media, emails, blogs, news items, books, research papers, and Q&A forums has increased significantly. This unstructured or semi-structured text contains a huge amount of information. Information Extraction (IE) can extract meaningful information from text sources and present it in a structured format. The sub-tasks of IE include Named Entity Recognition (NER), Event Extraction, Relation Extraction (RE), Sentiment Extraction, Opinion Extraction, Terminology Extraction, Reference Extraction, and so on.

One way to represent information in the text is in the form of entities and relations representing links between entities. The Entity Extraction task identifies entities from the text, and the Relation Extraction (RE) task can identify relationships between those entities. Many NLP applications can benefit from relational information derived from natural language, including Structured Search, Knowledge Base (KB) population, Information Retrieval, Question-Answering, Language Understanding, Ontology Learning, etc. This survey covers (1) basic concepts of Relation Extraction; (2) various Relation Extraction methodologies; (3) Deep Learning techniques for Relation Extraction; and (4) different datasets that can be used to evaluate the RE system.

1. Introduction

1.1. Introduction to Relation Extraction

Information Extraction (IE) extracts information from unstructured natural language text and represents it in a structured format (Jurafsky and Martin (2009)). IE can be used to construct a large-scale knowledge base (KB), which can be used by many downstream NLP applications to make knowledge-aware decisions. IE systems can extract various types of structures, such as entities, adjectives describing entities, relations, events, sentiments, opinions, and higher-level structures such as tables, lists, and ontologies (Sarawagi (2008)). This survey primarily focuses on the Relation Extraction (RE) sub-task of IE. RE extracts information from raw text and represents it in the form of a semantic relation between entities. For example, consider this sentence: “Sardar Patel was born in Nadiad.”. From this sentence, the RE system outputs a tuple (Sardar Patel, birth_place, Nadiad) representing a binary relation, where “Sardar Patel” and “Nadiad” represent two entities, and “birth_place” describes the relationship between these two entities. Many approaches to RE uses already extracted named entities and identifies relation between those entities using heuristics or machine learning approaches. Recent approaches tries to solve problems of NER (Named Entity Recognition) and RE in joint fashion.

1.2. History of Relation Extraction

RE dates back to the 1980s and has evolved since then. The Message Understanding Conference (MUC) was organised seven times between 1987 and 1998 (MUC-3 (1991), MUC-4 (1992), MUC-5 (1993), MUC-6 (1995), MUC-7 (1998)), which contributed significantly to the development of Relation Extraction technology. In each MUC, sample messages and instructions on the type of relation to be extracted were given to participating groups. The RE systems developed by participating groups were evaluated using text collection in that domain, and the results were scored according to certain metrics. Introduced in MUC-6 in 1995, the task of Named Entity Recognition (NER) involves identification and categorization of key information such as entity names (person and organization), locations, numerical expressions, email, date, and time (Grishman and Sundheim (1996)). In 1998, MUC-7 released a Template Relations (TR) task, which involves the identification of three simple relations: *location_of*, *employee_of*, and *product_of* (Marsh and Perzanowski (1998)). Another major contribution of the MUC7 conference was the introduction of evaluation indexes for Relation Extraction: Precision, Recall and F_1 -score. Table 1 shows the summary of MUC conferences.

Traditional IE was used extract pre-defined sets of relations from small homogeneous corpora specific to a domain. These approaches are

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Table 1
Details of MUC Conferences (Rodriguez (2021)).

Conference (Year)	Text Source	Topic	Task
MUC-1 (1987)	Military Reports	Fleet Operations	Open ended
MUC-2 (1989)	Military Reports	Fleet Operations	IE from the given template
MUC-3 (1991)	News Reports	Terrorist activities in Latin America	IE from the given template
MUC-4 (1992)	News Reports	Terrorist activities in Latin America	IE from the given template
MUC-5 (1993)	News Reports	Corporate Joint Ventures, Microelectronic production	IE from the given template
MUC-6 (1995)	News Reports	Negotiation of Labor Disputes and Corporate Management Succession	Named entity extraction, Coreference resolution and Scenarios
MUC-7 (1998)	News Reports	Airplane crashes, and Rocket/Missile Launches	Named entity extraction, Coreference resolution, Event Extraction and Scenarios

Table 2
Traditional IE vs Open IE.

	Traditional IE	Open IE
Input	Text Corpus + predefined set of relations	Text Corpus
Relations	Set of relations needs to be defined in advance	No need to define relations in advance
Extraction	Only predefined set of relations	All possible relations from the corpus
Corpus	Homogeneous Corpus	Heterogeneous Corpus

used to specify target relations and sets of patterns. The patterns are either hand-crafted or learned from hand-labelled training examples. These types of systems are called Rule-based RE systems (Riloff (1993), Appelt et al. (1993), Brin (1998), Muslea (1999), Ciravegna (2001), Maynard et al. (2001), Agichtein and Gravano (2000), Jayram et al. (2006), Shen et al. (2007)). They have the major disadvantage that if we move to a new domain, it requires redefining a new set of target relations as well as a new set of extraction patterns or new hand-labelled training examples. This disadvantage causes these approaches to rely heavily on human involvement.

To reduce human effort in RE, Banko et al. (2007) proposed a new relation extraction paradigm: *OpenIE*. In *OpenIE*, a user need not define a new target set of relations and extraction patterns in advance. It can extract all the sets of target relations automatically from the domain-independent heterogeneous corpora such as the web. Table 2 shows comparison between traditional IE and Open IE. Banko et al. (2007) developed the first *OpenIE* based RE system, *TEXTRUNNER*, which automatically extracted relations from the web. *TEXTRUNNER* inspired the development of various *OpenIE* based RE systems such as *WOE* (Wu and Weld (2010)), *REVERB* (Fader et al. (2011)), *OLLIE* (Mausam et al. (2012)), *ClausIE* (Del Corro and Gemulla (2013)), *Stanford OpenIE* (Angeli et al. (2015)), *OpenIE4* (Mausam (2016)), *PropS* (Stanovsky et al. (2016)), *OpenIE5* (Saha et al. (2017)), *Graphene* (Cetto et al. (2018)), *OpenIE6* (Kolluru et al. (2020)). These *OpenIE* systems learn extraction patterns automatically from labelled data or manually defined patterns and then use these learned patterns to extract relations from text corpora.

In recent years, Deep Learning (DL) has excelled in the fields of Computer Vision (CV) and Natural Language Processing (NLP). Many different Deep Neural Networks (DNNs) have been developed by researchers, including Recurrent Neural Networks (RNN), Convolution Neural Networks (CNN), Graph Neural Networks (GNN), and Recursive Neural Networks. With DL-based systems, several NLP applications, such as question answering, machine translation, sentiment analysis, text summarization, etc., have attained state-of-the-art performance. Many researchers have employed CNN (Liu et al. (2013), Zeng et al. (2014), Nguyen and Grishman (2015), Zeng et al. (2015), Lin et al. (2016), Jiang et al. (2016) and RNN Zhang and Wang (2015), Xu et al.

(2015), Zhang et al. (2015), Zhou et al. (2016), Xu et al. (2016), Qin et al. (2017), Stanovsky et al. (2018), Zhan and Zhao (2020)) for RE tasks. DL based methods have given significant improvements over previous non-DL methods.

1.3. Previous surveys

Bach and Badaskar (2007) presented one of the first surveys on RE approaches, but it does not reflect recent advances in the field. Another survey on RE Sarawagi (2008) covered a comprehensive study of RE techniques. Other recent surveys (Pawar et al. (2017), Zhang, Chen, et al. (2017)) cover supervised, semi-supervised, unsupervised, and distant supervision methods of RE. Another survey by Niklaus et al. (2018) describes various approaches for *OpenIE*. Most recent surveys (Kumar (2017), Wang et al. (2022), Nasar et al. (2021)) include detailed studies of various DL-based methods for RE.

In this survey, we have covered a comprehensive study of RE systems with a predominant focus on the DL-based RE approaches. We classified various techniques into two categories: traditional approaches, which were popular in the pre-DL era, and DL-based approaches. The remainder of the survey is structured as follows: Section 2 introduces various traditional methods of RE methods. Section 3 describes various DL-based methods of RE. Section 4 discusses various datasets available for the RE task. Section 5 and 6 covers progress of Relation Extraction systems in specific domains and low resource languages. Finally, Section 7 covers directions for future work in this field.

1.4. Evaluation metrics

Three common metrics are used to evaluate the performance of RE systems: Precision, Recall, and F_1 -score. Precision is defined as the percentage of true positive samples among the positive samples classified by the model. Recall is calculated as the division of positive samples correctly identified as positive and the total number of positive samples. We can't have precision and recall high at the same time. If we increase precision, then it will reduce recall, and if we increase recall, then it will reduce precision, which is called the precision-recall tradeoff. For this reason, the F_1 -score is used to measure a test's accuracy. An F_1 -score is used to compare the performance of two models. It combines precision and recall in the same equation by taking their harmonic mean. Researchers have proposed many benchmarks, such as Benchmarks WiRe57 (L  chelle et al. (2019)), BenchIE (Gashteovski et al. (2021)) and CaRB (Bhardwaj et al. (2019)) for the evaluation of RE systems.

$$\text{Precision} : P = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}}$$

$$\text{Recall} : R = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}}$$

$$F - \text{Measure} : F_1 = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

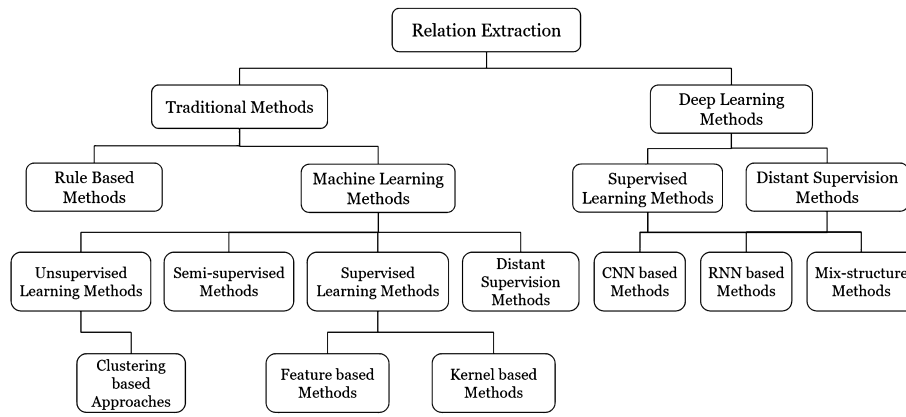


Fig. 1. Categories of Relation Extraction Methods.

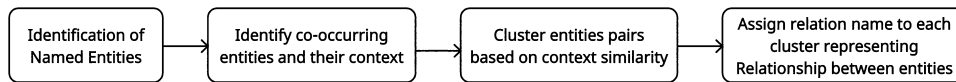


Fig. 2. Clustering based approach for Unsupervised Learning.

2. Traditional methods of RE

As shown in the Fig. 1, the existing RE methods can be categorised into two main types: Traditional methods and Deep Learning based methods. Traditional methods used either rule based techniques or machine learning techniques to extract predefined set of relations from a corpus. Traditional methods can be further categorised in Rule-based methods and Machine Learning-based methods. Machine learning-based methods can be further classified into four types: (1) Unsupervised methods, (2) Supervised methods, (3) Semi-supervised methods and (4) Distant Supervision methods.

2.1. Rule based methods

These methods are also called hand-built pattern methods. These types of methods define a set of extraction patterns for the predefined set of relations. Then these extraction patterns are matched with the text. If a pattern is matched, then a relation corresponding to the pattern is found in the text. (Riloff (1993), Appelt et al. (1993), Agichtein and Gravano (2000), Jayram et al. (2006), Shen et al. (2007), Fundel et al. (2006), Nebhi (2013)) have used ruled based methods to extract relations from the text. Table 3 shows examples of rules to extract hyponymy relations from text.

Rule-based methods require domain expertise and linguistic knowledge to define extraction patterns. These methods are domain specific, where the structure of the document is fixed and target relations are predefined. If we move from one domain to another domain, then we need to redefine a new set of target relations and extraction patterns. For this reason, rule-based methods require a lot of manual labour and they cannot be used for heterogeneous corpora.

2.2. Unsupervised methods

Unsupervised methods do not require any labelled data. Most unsupervised RE methods use a clustering-based approach. One of the earliest clustering-based unsupervised RE approaches was proposed in Hasegawa et al. (2004). They have used NE tagger to extract entities so that it can focus only on relations with those named entities mentioned. Fig. 2 shows steps of unsupervised learning method: (1) Identification of named entities in the text corpus, (2) Identify co-occurring named entities and their context, (3) Cluster entity pairs based on context similarity and (4) Assign semantic relation name to each cluster.

Table 3

Examples of Patterns for Hyponyms (Hearst (1992)).

Pattern	Input sentence	Extracted relations
<i>such NP as {NP,}*</i>	... works by such authors as Herrick, Goldsmith, and Shakespeare.	Hyponym ("author", "Herrick"), Hyponym ("author", "Goldsmith"), Hyponym ("author", "Shakespeare")
<i>NP {, NP} * {,} or other NP</i>	Bruises, wounds, broken bones or other injuries ...	Hyponym ("bruise", "injury"), Hyponym ("wound", "injury"), Hyponym ("broken bones", "injury")

K-means clustering based approach presented in Chen et al. (2005) uses feature selection mechanism based on frequency and entropy to identify important context words for various relation types. The results on the ACE corpus showed that the model is useful to extract subset of features and to estimate number of context clusters. Hierarchical clustering based approach Sekine (2005) uses keyword identification for relation extraction. First it identifies keywords and then group similar keywords into one collection. After that it links those collections with similar named entities. Another hierarchical clustering based approach WEBRE (Min et al. (2012)) uses entity similarity graph and hypernymy graphs for finding semantic similarity. WEBRE works in two phases: in first phase it identifies semantic classes representing large set of relations and then in second phase it groups similar relations together.

2.3. Supervised methods

Supervised methods require a large amount of training data, which is annotated with a set of entities and relations. It uses training data to train a classifier, which will extract relations from the test data. There are two types of supervised methods: Feature-based methods and Kernel-based methods (Pawar et al. (2017)). In feature-based approaches (Rink and Harabagiu (2010), Kambhatla (2004), Zhou et al. (2005)), a set of features is generated for each relation in the training data, and a classifier is trained to extract a new relation instance. Some lexical, syntactic, and semantic features are described in (Kambhatla (2004)). The selection of features affects the performance of a feature-based supervised RE system. Unlike feature-based methods, kernel-based methods (Bunescu and Mooney (2005b, 2005a), Zelenko et al. (2003), Culotta and Sorensen (2004), Zhang et al. (2008), Zhang (2004)) do not use preprocessing steps to design features. In kernel-based approaches, kernel functions are used to determine similarity

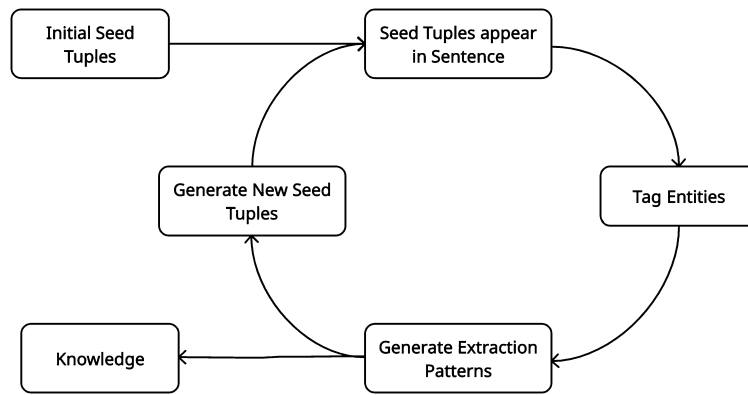


Fig. 3. Semi-Supervised Learning method for Relation Extraction.

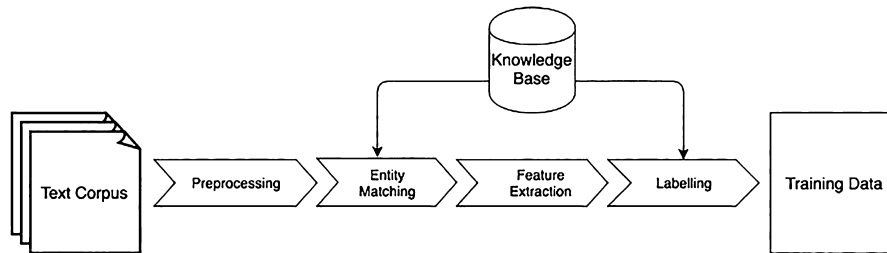


Fig. 4. Pipeline used for generating training data using distant supervision (Smirnova and Cudré-Mauroux (2018)).

between two relation instance representations, while Support Vector Machine (SVM) is used for classification. The performance of a kernel-based RE system is determined by the design of kernel functions.

2.4. Semi-supervised methods

Generating data for supervised methods of RE requires cost, effort, and time. Supervised methods generate labelled data automatically using bootstrapping algorithms. There are two advantages to this approach: (1) reduce the effort required to generate labelled data; (2) make use of freely available unlabelled data.

The bootstrapping algorithm requires a large amount of unlabelled data and a small number of seed instances of the desired relation type. For example, to extract relation *CapitalOf*, the seed examples (*New Delhi, India*), (*Canberra, Australia*) and (*London, England*) can be used to learn an extraction pattern. With these seed instances as input, the bootstrapping algorithm should be able to extract similar relations with entity pairs such as (*Paris, France*). Fig. 3 shows the model to extract patterns and seed tuples using semi-supervised learning method. KnowItAll (Etzioni et al. (2004)), TextRunner (Banko et al. (2007)), OLLIE (Mausam et al. (2012)), etc. are using a semi-supervised approach for RE.

2.5. Distant supervision methods

This is also called weakly supervised methods or knowledge based methods. Mintz et al. (2009) proposed a distant supervision method in which training data is automatically generated by aligning text with a Knowledge Base (KB). This eliminated the problem of hand-labelling of training data. Distance learning is based on the assumption, that if two entities with a relationship appear in KB, then all phrases that mention these two entities can express that relation.

In this way Distant supervision uses a knowledge base, such as Freebase, to extract relationships between two things. When the same entity pair appears in both a sentence and a KB, the sentence is heuristically associated with the matching relation from the KB. For example, consider a sentence, “Bill Gates is the founder of Microsoft.”. If person “Bill Gates” and organization “Microsoft” appear as triple (*entity1: Bill Gates,*

entity: Microsoft, relation: founder_of) in Freebase, then these two entities represent relation *founder_of*. Smirnova and Cudré-Mauroux (2018) used a pipeline as shown in Fig. 4 for generating training data using distant supervision. Other approaches (Riedel et al. (2010), Takamatsu et al. (2012), Zeng et al. (2014), Qin et al. (2018), Chen and Manning (2014), Quirk and Poon (2017)) use Distance supervision for extracting relations from the text.

Table 4 shows the comparison of traditional Relation Extraction systems, their features and performance.

3. Deep learning methods

Deep Neural Networks (DNNs) (Bengio (2009)) have gained popularity over the past decade for a wide range of applications, including natural language processing (NLP), computer vision (CV), and speech recognition. DNN architectures based on Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long short-term Memory (LSTM), Gated Recurring Unit (GRU), Bidirectional Encoder Representations from Transformers (BERT) are used for different NLP tasks such as POS tagging (Gopalakrishnan et al. (2019), Srivastava et al. (2018), Kumar et al. (2018)), machine translation (Wu et al. (2016), Wang et al. (2019), Gehring et al. (2017)), question answering (Zhu et al. (2018), Lei et al. (2018), Liu et al. (2020)), semantic role labelling (He et al. (2017), Zhou and Xu (2015), Marcheggiani and Titov (2017)), dialogue generation (Li et al. (2016), Miranda and Kessaci (2020), Zhao and Eskenazi (2016)), text generation (Chen et al. (2020), Raffel et al. (2020), Li et al. (2021)), sentiment analysis (Yu et al. (2019), AL-Smadi et al. (2019), Zhao et al. (2018)), automatic summarization (Zhang et al. (2019), See et al. (2017), Liu and Lapata (2019), Nallapati et al. (2016), Cai et al. (2019)) etc.

3.1. CNN based methods

Convolution Neural Networks (CNNs) have achieved state-of-the-art performance in image processing tasks. CNN can be employed in many NLP applications to extract word and sentence level features. Input sentence x with m words is defined as sequence of word vectors

Table 4
Comparison of Traditional RE systems.

Technique	Reference	Feature Sets Model Properties	Dataset/Source	Evaluation Metrics		
				P	R	F
Rule based (AutoSlog)	Riloff (1993)	Rule based heuristics	MUC-4	43.00	56.00	50.51
Rule based (FASTUS)	Appelt et al. (1993)	Pattern matching based on Finite state automata	Blind test	44.00	55.00	-
Rule based (Snowball)	Agichtein and Gravano (2000)	Pattern matching, Automatic Pattern Generation	North American News Text Corpus	76.00	45.00	-
Rule based (RelEx)	Fundel et al. (2006)	Rule matching, Conjunction, Negation	LLL 2005 workshop	68.00	83.00	75.00
Rule Based	Nebhi (2013)	Rule matching, Syntactic parser, DBPedia	Quaero Newss	75.50	62.10	68.10
Unsupervised Learning	Hasegawa et al. (2004)	Hierarchy NER, Complete Linkage clustering	NYT	77.50	78.50	77.50
Unsupervised Learning	Chen et al. (2005)	Unsupervised Feature Subset Selection, K-means clustering	-	-	-	-
Unsupervised Learning	Sekine (2005)	Hierarchical Clustering, Information chunking	News	Accuracy:79.5		
Unsupervised Learning (WEBRE)	Min et al. (2012)	Hierarchical Clustering, Entity similarity graphs, hypernymy graphs	Cluewebset 2009	89.70	68.40	77.60
SVM	Zhou et al. (2005)	Lexical, Semantic, Syntactic, External Lexicon	ACE 2003	77.20	60.70	68.00
SVM	Rink and Harabagiu (2010)	Contextual, Lexical, Nominal affiliation, pre-existing relations features	SemEval 2010 Task-8	82.25	82.28	82.19
MEMM	Kambhatla (2004)	Lexical, Semantic and Syntactic features	ACE 2003	63.50	45.20	52.80
SVM	Bunescu and Mooney (2005b)	Subsequent kernels	ACE 2000	73.90	35.20	47.70
SVM	Zelenko et al. (2003)	Kernels and voted perceptron	200 newswire and publications	82.70	91.30	86.00
SVM	Culotta and Sorensen (2004)	Tree Kernel	ACE 2003	70.3	26.30	38.00
SVM	Zhang et al. (2008)	Tree Kernel	ACE 2003	76.10	68.40	72.10
Bootstrapping with SVM	Zhang (2004)	Radial Bias Kernel	Self annotated	63.20	61.50	60.30
SVM	Bunescu and Mooney (2005a)	Shortest Path Dependency Kernel	ACE 2000	65.50	43.80	52.50
CRF	Culotta et al. (2006)	Relational pattern features, word, external features	Wikipedia pages	73.40	56.10	63.60
Distant Supervision	Riedel et al. (2010)	SampleRank, CRF, FreeBase	New York Times dataset	87.00	-	-
Distant Supervision	Takamatsu et al. (2012)	Logistic Regression, Freebase, EM	Wikipedia articles	78.20	68.20	66.70

SVM - Support Vector Machines, MEMM - Maximum Entropy Markov Models.
Some of the results are referenced from Nasar et al. (2021).

$x = w_1, w_2, \dots, w_m$, where w is a d -dimensional word vector ($w_i \in \mathbb{R}^d$). For Convolution layer kernel with window size l , vector for the i^{th} window $q_i \in \mathbb{R}^{(d \times l)}$ is formed by concatenating input vectors for that window,

$$q_i = w_{i:i+1-l}; \quad (1 \leq i \leq m+1-l)$$

The output of a single convolution kernel for the i^{th} window with a weight vector $W \in \mathbb{R}^{(d \times l)}$, a bias $b \in \mathbb{R}$ and the activation function f is computed as,

$$p_i = f(W'q_i + b)$$

The output of the convolutional kernel p would be of the shape $p \in \mathbb{R}^{(m-l+1)}$. If a convolutional layer can consist of d_c convolutional kernels, it would make the output of the convolutional layer of the shape $\mathbb{R}^{d_c \times (m+1-l)}$ (Kumar (2017)).

3.1.1. CNN based supervised IE

The initial work on relation extraction using deep learning was based on a supervised learning paradigm with a hand-labelled training corpus. The model considers the RE task as a multi-class classification problem, where the model assigns a relation class to a sentence containing the mentioned entity pair.

Simple CNN based model (Liu et al. (2013)) One of the earliest models by Liu et al. (2013), used a CNN based architecture for relation extraction. It builds end-to-end neural network architecture with three main building blocks: a input layer, a convolution layer, and a classic neural network layer. The input layer implements a lookup table to transform input sentences into word vectors using lexical features and a synonym dictionary. The convolution layer is implemented using a sequential kernel, which maps word vectors from the input layer to a new vec-

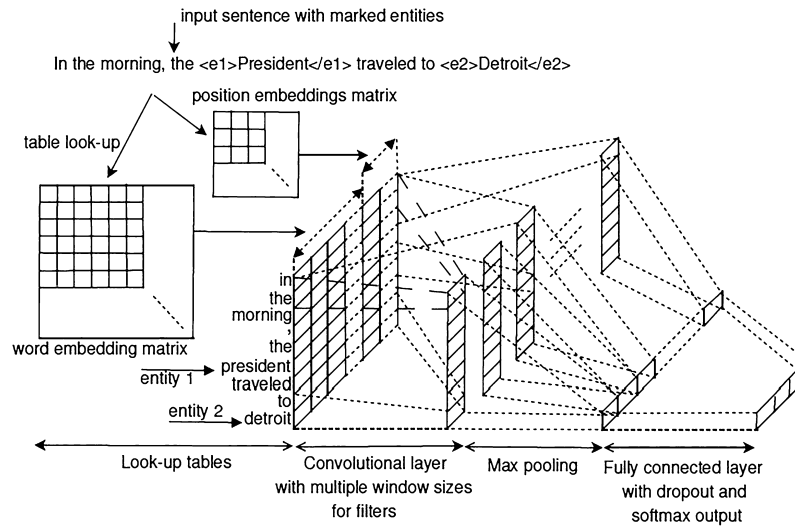


Fig. 5. Convolutional Neural Network with Multi window sizes and max-pooling for Relation Extraction (Nguyen and Grishman (2015)).

tor space. The output of convolution layer is fed to the neural network with softmax to compute classification probability. The model achieved 83.8% F-measure by out-performing the state-of-the-art model (Typed Dependency Kernel method - Reichartz et al. (2010)) of that time on the ACE dataset.

CNN model with max-pooling (Zeng et al. (2014)) This model used CNN to encode sentence level features. Unlike Liu et al. (2013), it used word embedding pre-trained on large text corpora. It introduced the concept of Position Embedding (PE), which became standard for all subsequent RE Deep Learning models. This model also took into account lexical data such as information on the nouns in the sentence and the nouns' WordNet hypernyms. Another significant contribution of this approach was the introduction of a max-pooling layer over the output of the convolutional network. The output of the convolutional layer $Z \in \mathbb{R}^{d_c \times (m-l+1)}$ is dependent on the size of the input sentence m . Max-pooling operation collapse convolution layer output Z into $Z' \in \mathbb{R}^{d_c}$, by capturing most important feature in each dimension, and thus making Z' independent of input sentence length. Their research also showed that the use of PE improved their F1-score by nearly 9%. In 2014, the model outperformed SVM and Max Entropy-based models that incorporated various lexical characteristics and attained state-of-the-art results in the RE task on the SemEval-2010task8 dataset.

CNN model with multiple window size filter (Nguyen and Grishman (2015)) Earlier work Zeng et al. (2014) used fixed window size filters for RE, which mainly focused on local features and ignored global features. Another research Riedel et al. (2010) proposed the use of multi-window size filters to incorporate more structured information to the model. The model initialised word embedding with pre-trained word vectors and trained with word2vec (Mikolov et al. (2013)), which gives them significant performance improvements over random word vectors and static-word2vec vectors. Fig. 5 depicts the model architecture, which consists of word embeddings and positional embeddings followed by a convolution layer with filters with multiple window sizes. The output of the convolutional layer is fed to the max-pooling layer followed by the softmax layer. The convolution layer uses kernels with varying window sizes to capture a wider range of n-gram features. After experimenting with different iterations, they discovered that kernels with window lengths of 2-3-4-5 give the best results.

CNN model with classification by ranking (Dos Santos et al. (2015)) Instead of using the cross entropy loss function, the CR-CNN (Classification by Ranking CNN) model used the pairwise ranking loss function.

A model trained using the ranking loss function, gives a higher score for correct classification and a negative score for incorrect classification. According to the authors, using word embeddings as the only input feature is adequate to produce state-of-the-art performance if the text between two target entities is considered.

3.1.2. CNN based distant supervision IE

As mentioned in section 2.5, the idea behind distant supervision is that if two entities participate in a relation, any sentence containing those two entities might express that relation (Mintz et al. (2009)). Riedel et al. (2010) argued that the preceding assumption is excessively strong, resulting in a noisy label, and that it should be loosened. They proposed another assumption, *expressed-at-least-once*: If there are two entities involved in a relationship, at least one statement referencing both of them can be used to express that relationship. Riedel et al. (2010) suggested a new approach, where RE tasks can be considered as Multi-Instance Learning (MIL) problems. MIL is a type of supervised learning where training examples are divided into sets, called bags, and a label is applied to the entire bag. With multi-instance learning approach and assumption suggested by Riedel et al. (2010), RE task can utilize large training data generated using distant supervision, while reducing noise in labels.

Piecewise CNN (PCNN) (Zeng et al. (2015)) The PCNN model uses a distance learning paradigm with multi-instance learning to extract relations. It proposed a similar architecture as Zeng et al. (2014) and Nguyen and Grishman (2015), but with the addition of a piecewise max-pooling operation to extract important features automatically. According to the authors, a single max-pooling operation is insufficient for capturing all the structural information in a sentence. A sentence can be broken up into three segments depending on the positions of two things. They employed piecewise max-pooling on each segment of the convolution layer's output to extract the maximum value in each segment rather than a single maximum value for the entire sentence. It gives better performance by capturing more structural and latent information from the sentence. The architecture of PCNNs is shown in Fig. 6.

Due to the underlying assumption, one of the drawbacks of this model is that it selects the highest probability sentence for each entity pair during training and prediction. Thus, it neglects a large amount of useful data and information expressed in other sentences in the bag. Even though all sentences in the bag may not represent correct relations between entity pairs, using only a single sentence from the pair is a hard constraint. This drawback is addressed in subsequent work. The

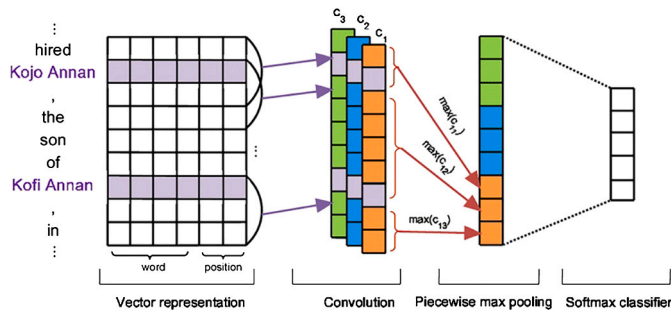


Fig. 6. The architecture of PCNNs used for distant supervised relation extraction (Zeng et al. (2015)).

PCNNs model with multi-instance learning has outperformed three traditional methods: the traditional distance learning model (Mintz et al. (2009)), multi-instance learning models proposed in *MultiR* (Hoffmann et al. (2011)) and *MIML* (Surdeanu et al. (2012)).

Piecewise CNN with selective attention (PCNN + ATT) (Lin et al. (2016)) To address the shortcoming of the previous model by Zeng et al. (2015), Lin et al. (2016) used an attention mechanism over all sentences in the bag for multi-instance problem. The model computes selective attention for each sentence in the bag, based on how well the input sentence matches the predicted relation. Sentence level attention across multiple instances is intended to reduce the weight of noisy sentences, thereby resolving the problem of incorrect labelling. PCNN method (Zeng et al. (2015)) is a special case of the selective attention method where sentence level attention is set to 1 for the highest probability sentence and 0 for all other sentences in the bag. This model, with multi-instance learning with selective attention, achieves significant performance improvement over PCNN by gathering useful information over multiple sentences in the bag.

Multi-instance multi-label CNN (MIMLCNN) (Jiang et al. (2016)) The distance supervision paradigm expressed in Zeng et al. (2015) selects only one sentence to represent an entity pair during the training phase, which wastes information in neglected sentences. MIMLCNN relaxes the assumption as follows: a relation held between two entities can be either expressed explicitly or inferred implicitly from all sentences that mention these two entities. MIMLCNN uses cross-sentence max-pooling to take advantage of this assumption and utilise information available in all sentences. By using cross-sentence max-pooling, each feature in the final feature vector can derive from the sentence that is most significant for that feature alone, as opposed to the feature vector as a whole coming from the overall most significant sentence.

Existing neural networks adopt multi-instance learning with a single label, where a single relation label is assigned between entity pairs. In distant supervision, there can be instances, where multiple relations exist between entity pairs. For example, (*Elon Musk, Founded, Tesla*) and (*Elon Musk, CEO of, Tesla*) are both valid relations between the same entity pair (*Elon Musk, Tesla*), which may be supported by different sentences. The MIMLCNN model modifies neural network architecture for distant supervision to solve multi-label learning problems by using a sigmoid activation function instead of a softmax activation function, which allows the model to calculate the probability of each relation class independently, rather than a probability distribution over relation classes. The authors showed that MIMLCNN is able to increase the performance of PCNN and CNN models using cross-sentence max-pooling multi-label learning.

Table 5 shows summary of CNN based models and their performance on various datasets. It is evident from the table that all CNN based RE models outperforms traditional RE models. Models trained on NYT dataset achieves less performance compare to SemEval-2010 Task 8 dataset, as there are very less number of examples (500 examples) in

NYT dataset compare to SemEval-2010 Task 8 dataset (approx 10,000 examples), which shows more number of training examples helps model achieve better performance.

3.2. RNN and LSTM based methods

CNNs are very good at learning local patterns, but they lack the capacity to capture long-distance dependency and time sequence information. Various RNN, LSTM, and GRU-based models that are better suited for sequence modelling can be employed for NLP applications like Relation Extraction.

Simple RNN (Zhang and Wang (2015)) The RNN based model proposed in Zhang and Wang (2015) tries to model sequential data and learns long-distance patterns using bi-directional RNN. Fig. 7 shows the architecture of the model. Each word in the sentence is represented by a pre-trained word vector in the model's embedding layer. The next layer in the model employs bi-directional RNN to model word-level features and long-distance dependency. The bi-directional architecture helps the model predict semantic meaning by looking at past and future words in the sentence. In the final layer, the model used a max-pooling approach similar to CNN to extract sentence-level features by merging local features. In RNNs, the semantic meaning of a sentence is learned by sequentially processing all the words in the sentence, so the final vector at the end of the sentence represents sentence level features. This accumulation based approach to learning sentence level features may not be suitable as it can forget long-term information.

SDP-LSTM (Xu et al. (2015)) RNN based models suffer from vanishing and exploding gradient problems, which can be overcome using LSTM (Hochreiter and Schmidhuber (1997)). In this model, the first Shortest Dependency Path (SDP) is extracted using Stanford parser¹ (Chen and Manning (2014)). In addition to SDP information, four distinct forms of information are sent to LSTM units via four different channels: words, Part of Speech (POS) tags, WordNet hypernyms, and grammatical relations. Finally, the outputs of all four channels are concatenated and given to the softmax output layer for final classification. SDP provides valuable information about the target relation, so it can be a useful resource for relation classification.

BLSTM (Zhang et al. (2015)) The architecture of the model is similar to that proposed in Zhang et al. (2015) with bi-directional LSTM. The bi-directional long short-term memory network (BLSTM) allows input to flow in both directions and is capable of utilising information from both directions, helping the model obtain better sentence level representation. The model uses various features such as word embedding, POS, NER, WordNet hypernyms, positional features (PF), and dependency features as input to the model. The authors demonstrated that it is possible to achieve state-of-the-art performance using only word embedding as an input feature to BLSTM. Additional lexical and positional features can improve the performance of the model.

Att-BLSTM (Zhou et al. (2016)) In order to extract the most significant semantic information from the sentence, the authors proposed a BLSTM model with an attention mechanism. Fig. 8 shows the architecture of the BLSTM model with an attention mechanism. This model, which uses simple features like word embedding and positional indications, shows similar performance to the state-of-the-art BLSTM model (Zhang et al. (2015)), which extracts initial features using NLP toolkits and lexical resources.

DRNN (deep recurrent neural network) (Xu et al. (2016)) Previous neural network models discussed in this section are designed using a

¹ Stanford Parser: <http://nlp.stanford.edu/software/lex-parser.shtml>.

Table 5
Comparison of CNN based RE systems.

Model	Reference	Model Properties	Dataset	Word Embeddings	Evaluation Metrics		
					P	R	F
CNN	Liu et al. (2013)	Supervised learning, Fix-window size filter, Synonym coding	ACE 2005	One hot	83.7	83.9	83.9
CNN	Zeng et al. (2014)	Supervised learning, Lexical level features, Fix-window size filter, Max pooling, PE	SemEval-2010 Task 8	pre-trained	82.7	-	82.7
CNN	Nguyen and Grishman (2015)	Supervised learning, Multi-windows size filters, Max pooling, PE	SemEval 2010 dataset	word2Vec	-	-	82.8
CR-CNN	Dos Santos et al. (2015)	Supervised learning, Fixed filter size, Ranking Loss Function, PE	SemEval-2010 Task 8	word2vec	83.7	84.7	84.1
PCNN	Zeng et al. (2015)	Distant Supervision, Piecewise CNN, Multi-window size filter, PE	NYT	word2vec	78.3	-	-
PCNN + ATT	Lin et al. (2016)	Distant Supervision, Piecewise CNN, Fix-window size filter, Sentence-level Attention, PE	NTY	word2vec	72.2	-	-
MIMLCNN	Jiang et al. (2016)	Distant Supervision, Multi-window size filter, Piecewise max-pooling, PE	NYT	word2vec	72.0	-	-
CNN	Wang et al. (2016)	Multi-level Attention	SemEval-2010 Task 8	word2vec	88.0	-	88.0
PCNN	Qu et al. (2018)	Distant Supervision, Piecewise CNN, Word-level attention	NYT	word2vec	46.9	44.5	45.7
CNN	Li et al. (2018)	Multi-path CNN, Word and Sentence level attention	NYT	word2vec	77.0	-	-

PE: Positional Embeddings.

Some of the results are referenced from Nasar et al. (2021).

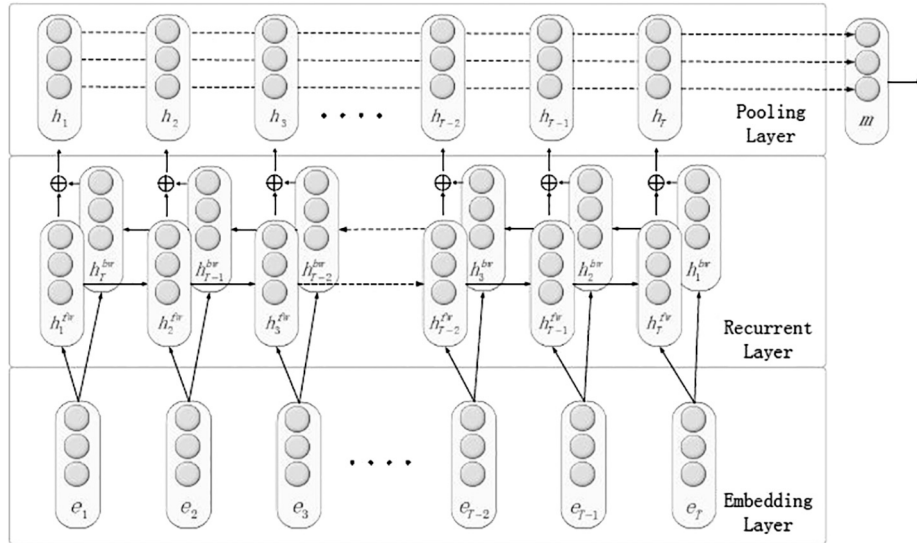


Fig. 7. Architecture of simple bi-directional RNN based model (Zhang and Wang (2015)).

shallow architecture with one layer of RNN or LSTM. Deep learning models with multiple hidden layers give better performance than shallow networks. With this idea in mind, DRNNs extends the SDP-LSTM (Xu et al. (2015)) model by adding multiple (up to four) hidden layers, allowing the model to capture different information at different levels of abstraction. The SemEval-2010 relation classification dataset consists of only a few thousand samples. A deep neural network with small training data is prone to overfitting problems. Authors tried to solve the

overfitting problem by adding directionality of the relation using data augmentation techniques. With deep architecture and data augmentation techniques, DRNNs achieved state-of-the-art performance on the SemEval-2010 Task 8 dataset.

EAtt-BiGRU (Qin et al. (2017)) Entity-pair-based Attention Bidirectional GRU (EAtt-BiGRU) model feeds entity pair information as prior knowledge to the attention layer, which allows the model to get adap-

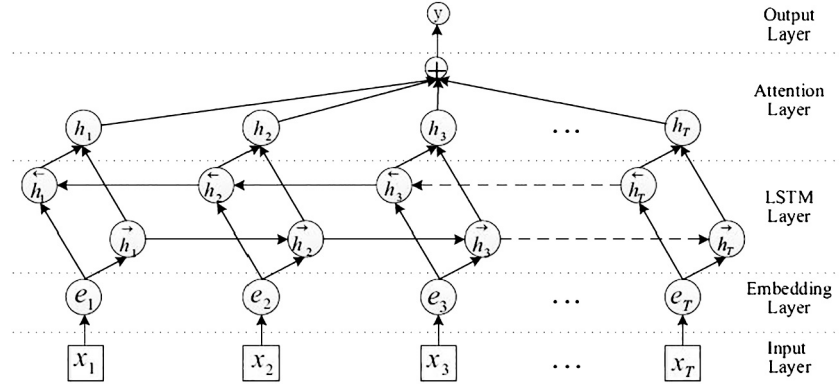


Fig. 8. Bidirectional LSTM model with Attention (Zhou et al. (2016)).

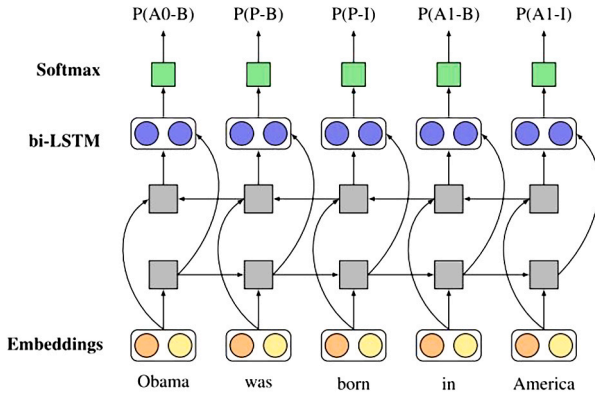


Fig. 9. RNNOIE model architecture (Stanovsky et al. (2018)).

tive attention weights. In contrast to Att-BLSTM (Zhou et al. (2016)), EAtt-BiGRU reduces computational costs by using bi-directional GRU (BiGRU) rather than bi-directional LSTM (BLSTM). Similar to Att-BLSTM (Zhou et al. (2016)), this model also uses word embedding and positional embedding as input. With this input information and prior knowledge about entity pairs, EAtt-BiGRU is able to achieve better performance than Att-BLSTM (Zhou et al. (2016)).

RNNOIE (Stanovsky et al. (2018)) The bidirectional LSTM-based model proposed by Stanovsky et al. (2018), represents the RE task as a sequence tagging problem. The model is fed word embedding as well as POS information from the token and predicate's heads as input. The model generates an output sequence with custom BIO (Beginning-Inside-Outside) tags (Ramshaw and Marcus (1995), Tjong Kim Sang and Veenstra (1999)) representing entities and relation between entities. Fig. 9 depicts the architecture of the model. RNNOIE employed a bi-LSTM transducer, which consists of three layers of bi-LSTM cells and a softmax layer, to predict one of the possible BIO tags.

The paper also covers an OpenIE training corpus named All Word OpenIE (AW-OIE), which is derived from the Question-Answer Meaning Representation (QAMR) (Michael et al. (2018)) dataset by extracting predicates and arguments from question-answer pairs. The model is evaluated using the benchmarking suite Schneider et al. (2017), which includes previous evaluation datasets (WEB, NYT (Mesquita et al. (2013)), PENN (Xu et al. (2013)) and OIE2016 (Stanovsky et al. (2016))).

SpanOIE (Zhan and Zhao (2020)) Zhan and Zhao (2020) proposed the SpanOIE model for n-ary information extraction from an accurately annotated corpus. A span (i, j) is the part of the sentence that begins with the word i and ends with j . For a sentence $W = w_1, w_2, \dots, w_n$, set of all span is represented as $S = \{(w_i, \dots, w_j) | 1 \leq i \leq j \leq n\}$. SpanOIE model is divided into two sub-models: the Span model and the Argument model.

The span model employs a bi-LSTM network to derive span features, which are then fed into a feed forward network (FFN) and a Softmax layer to predict whether or not span is a predicate. Similar to the span model, the argument model also uses bi-LSTM to predict labels for arguments. Authors found that, OIE2016 (Stanovsky et al. (2016)) corpus contains some incorrect extractions, so they relabelled OIE2016 corpus to improve accuracy of the SpanOIE model. Table 6 shows comparison of various RNN based models and their performance.

3.3. Encoder-decoder/transformer based methods

Encoder-decoder based models were proposed in (Cho et al. (2014), Sutskever et al. (2014)). The encoder consists of multiple layers of LSTMs or GRUs, which receive a sequence of symbols as input and produce a fixed length vector, representing a compact representation of the input sequence. The decoder also consists of multiple layers of LSTMs or GRUs, which maps the output of the encoder to the target sequence. The encoder-decoder architecture with the attention mechanism introduced in (Bahdanau et al. (2015), Luong et al. (2015)) further enhanced performance by allowing the decoder to focus on the parts of the input sequence that are relevant to the target word. Vaswani et al. (2017) proposed Transformer architecture, which is based on an encoder-decoder architecture with self-attention.

Neural OpenIE (Cui et al. (2018)) Neural OpenIE is an encoder-decoder based framework, which treats RE task as sequence-to-sequence generation problem, where input is a sequence of tokens and output is a sequence of tokens with delimiters indicating entities and relation boundaries. As shown in Fig. 10, the Neural OpenIE architecture employs an encoder-decoder framework with attention and attention-based copying. In this architecture, there are 3-layers of LSTM network in both the encoder and decoder. The encoder receives text sequence of varying length as input and converts it into a hidden representation. The decoder takes the encoder's output and attention information as input and gives it to the LSTM network followed by softmax to generate the final output sequence. The model uses a copying mechanism to reduce the number of unknown words in the output sequence.

The Neural OpenIE system was able to perform better than various state-of-the-art systems, including Stanford OpenIE (Angeli et al. (2015)), OLLIE (Mausam et al. (2012)), PropS (Stanovsky et al. (2016)), ClausIE (Del Corro and Gemulla (2013)) and OpenIE4 (Mausam (2016)). A performance comparison of the above systems is shown in Fig. 11. The neural encoder-decoder architecture is good at identifying relation boundary, which helps it to achieve better performance than previous methods. Another encoder-decoder framework-based model (Nayak and Ng (2020)) employed bi-LSTM in both the encoder and decoder sides, as well as an attention and copy mechanism.

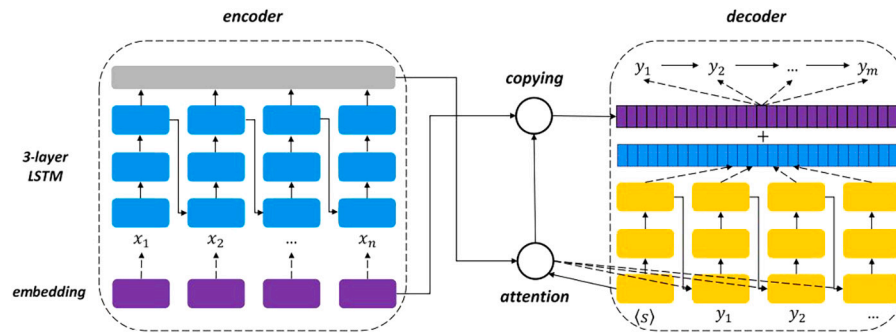
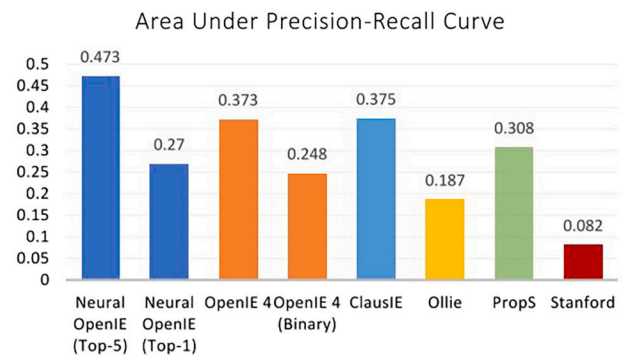
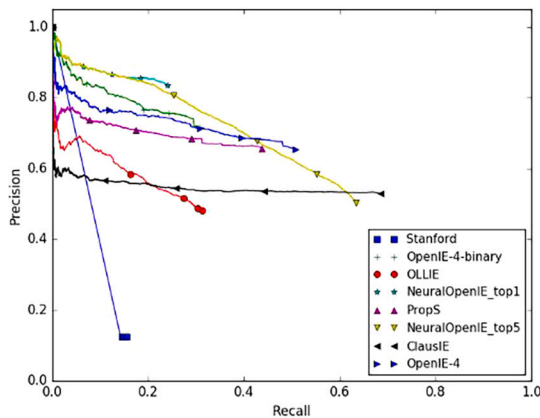
Transformer based OpenIE (Han and Wang (2021)) This paper proposed an Open IE system based on the transformer-architecture

Table 6

Comparison of RNN based RE systems.

Model	Reference	Model Properties	Word Embeddings	Dataset	Evaluation Metrics		
					P	R	F
MV-RNN	Socher et al. (2012)	POS tags, NER tags, Wordnet Hypernymy	pre-trained	SemEval-2010 Task 8	82.4	-	82.4
RNN	Zhang and Wang (2015)	Bi-directional RNN, Max-pooling, PI	word2vec	SemEval-2010 task 8	-	-	79.6
SDP-LSTM	Xu et al. (2015)	SDP, POS tags, Grammatical relations, Wordnet Hypernymy	word2vec	SemEval-2010 task 8	-	-	83.7
BLSTM	Zhang et al. (2015)	Bi-directional LSTM, POS tags, NER tags, Dependency parse tree, Wordnet Hypernymy	GloVe	SemEval-2010 task 8	-	-	83.6
Att-BLSTM	Zhou and Xu (2015)	Bi-directional LSTM, Attention	GloVe	SemEval-2010 Task 8 dataset	-	-	84.0
DRNN	Xu et al. (2016)	POS, grammatical relations, Wordnet Hypernymy, data augmentation	word2vec	SemEval-2010 Task 8	-	-	86.10
Bi-LSTM	Li et al. (2017)	Stacked LSTMs	pre-trained	PubMed abstracts	67.50	75.80	71.4
EAtt-BiGRU	Qin et al. (2017)	Bi-directional GRU, PE	word2vec	SemEval-2010 Task 8	-	-	84.7
RNNOIE	Stanovsky et al. (2018)	Bi-directional LSTM, POS embedding	GloVe	OIE2016 WEB NYT PENN	61.0 83.0 24.0 31.0	64.0 56.0 67.0 75.0	62.0 67.0 35.0 44.0
SpanOIE	Zhan and Zhao (2020)	Bi-directional LSTM, Spans Candidates Selection	GloVe	OIE2016	-	-	78.5

POS: Part of Speech, NER: Named Entity Recognition, PI: Positional Indicators, SDP: Shortest Dependency Path, PE: Position Embeddings.

**Fig. 10.** Encoder-decoder model architecture of Neural OpenIE system (Cui et al. (2018)).**Fig. 11.** Performance comparison of Neural OpenIE with other Open IE systems Cui et al. (2018).

(Vaswani et al. (2017)). The transformer model architecture is shown in Fig. 12. An encoder is stack of multiple identical layers, where each layer consists of two sub-layers. The first sub-layer is the self-attention layer, which allows the encoder to relate different tokens in the input sequence. The second sub-layer consists of Feed-Forward Neural Net-

work (FFN). The decoder consists of these two layers, and it also has a third layer with multi-head attention, which helps the decoder identify relevant tokens from the input sequence. A performance comparison of the transformer based model with various OpenIE systems is shown in Fig. 13.

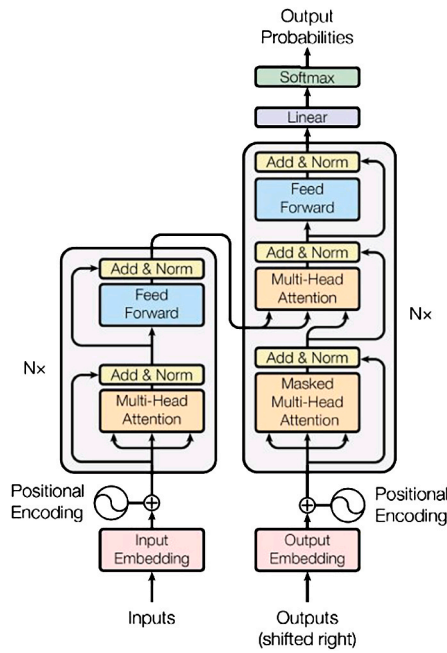


Fig. 12. Transformer model architecture (Vaswani et al. (2017)).

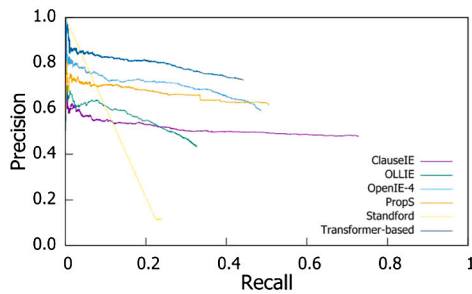


Fig. 13. Transformer model performance comparison with OpenIE systems (Han and Wang (2021)).

Multi²OIE (Ro et al. (2020)) Similar to RNNOIE (Stanovsky et al. (2018)), Multi²OIE treats RE task as sequence labelling problem. Multi²OIE uses BERT (Devlin et al. (2019)) combined with multi-head attention (Vaswani et al. (2017)) blocks to perform IE. The architecture of Multi²OIE shown in Fig. 14. In the first part, the predicate model uses BERT followed by FFN and a softmax layer to identify the predicate in the sentence. In the second part, the argument model uses BERT hidden state combined with predicate information and position embedding as input to the multi-head attention block, followed by FFN and softmax to predict arguments. Fig. 15 depicts a performance comparison of the Multi²OIE model with other OIE models on the Re-OIE2016 (Zhan and Zhao (2020)) and CARB (Bhardwaj et al. (2019)) datasets.

The authors used multilingual-BERT in the Multi²OIE model to perform multilingual OIE. The Multilingual-Multi²OIE model was trained using the Re-OIE2016 English dataset. The Re-OIE2016 dataset is translated into two target languages: Spanish and Portuguese, using Google² for evaluation. Table 7 shows the performance of the multilingual-Multi²OIE system on the English, Spanish and Portuguese languages. The Multilingual-Multi²OIE model's performance on the English version is lower compared to the Multi²OIE system, but it still gives better performance than previous sequence-labelling systems. The Spanish and Portuguese versions of the Multilingual-Multi²OIE system outperform rule-based multilingual systems ArgOE (Gamallo and Garcia (2015))

² <https://cloud.google.com/translate/>.

Table 7

Performance of Multilingual OIE systems on multilingual Re-OIE2016 dataset (Ro et al. (2020)).

Lang.	System	F1	PRECISION	RECALL
EN	ArgOE	43.4	56.6	35.2
	PredPatt	53.1	53.9	52.3
	Multi ² OIE	69.3	66.9	71.7
ES	ArgOE	39.4	48.0	33.4
	PredPatt	44.3	44.8	43.8
	Multi ² OIE	60.2	59.1	61.2
PT	ArgOE	38.3	46.3	32.7
	PredPatt	42.9	43.6	42.3
	Multi ² OIE	59.1	56.1	62.5

and PredPatt (White et al. (2016)) with training using only the English language.

4. Popular data sets

In the field of RE, a large-scale corpus with annotated tuples is required to train and evaluate RE systems. Most of the datasets are generated either by manual annotation, by crowd-sourcing, or derived from other datasets. In the pre-DL era, various datasets, such as WEB-500, NYT-500, NYT-222, PENN-100, ACE2005, were used for training RE systems. These datasets contained very few sentences with hand-annotated arguments and relations. DL-based IE systems are data intensive so they require large datasets containing high quality tuples with little or no noise to train the system. It can take a long time to manually annotate tuples for huge datasets, so researchers have converted certain datasets from previous NLP tasks. The list of Relation Extraction dataset is shown in Table 8.

WEB-500 The WEB-500 dataset is a very common dataset, which was developed for the evaluation of the TextRunner system (Banko et al. (2007)). It is a very noisy dataset, and it contains incomplete sentences, incorrect grammar, HTML-character encoding, and special characters in headlines.

ACE2005 The Automatic Content Extraction 2005 (ACE2005) dataset contains annotated events and relations in English, Arabic, and Chinese languages. It contains 599 English documents from various sources, such as news, forums, and telephone conversations. It contains 7 major relation types, with an average of approximately 700 examples per relation.

NYT-500 The NYT-500 dataset contains 500 sentences from the New York Times corpus, manually labelled with binary relations.

NYT-222 The NYT-222 (Mesquita et al. (2013)) dataset contains 222 sentences with manually tagged n-ary relations from the New York Times.

PENN-100 The PENN-100 (Xu et al. (2013)) dataset was developed for the evaluation of Tree Kernel method of the Open Relation Extraction. It contains 100 sentences from Penn Treebank with 51 manually annotated relations.

SemEval-2010 Task 8 The SemEval-2010 Task 8 dataset (Hendrickx et al. (2010)) is designed for the semantic relations classification task. There are 10717 relation instances in all, 8000 of which are training examples and 2717 of which are testing cases. It contains 9 relation classes and 1 other class, which are ordered relations. It would double the number of relations due to directionality, so effectively the dataset contains 19 relation classes ($9 \times 2 + 1$ for the other class).

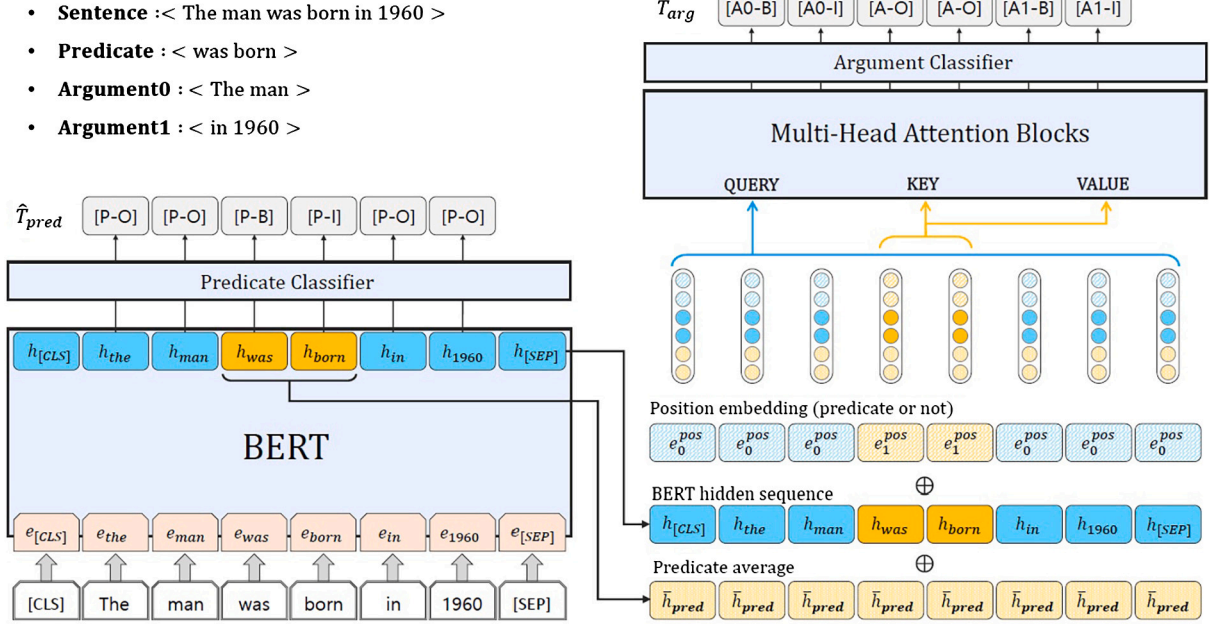
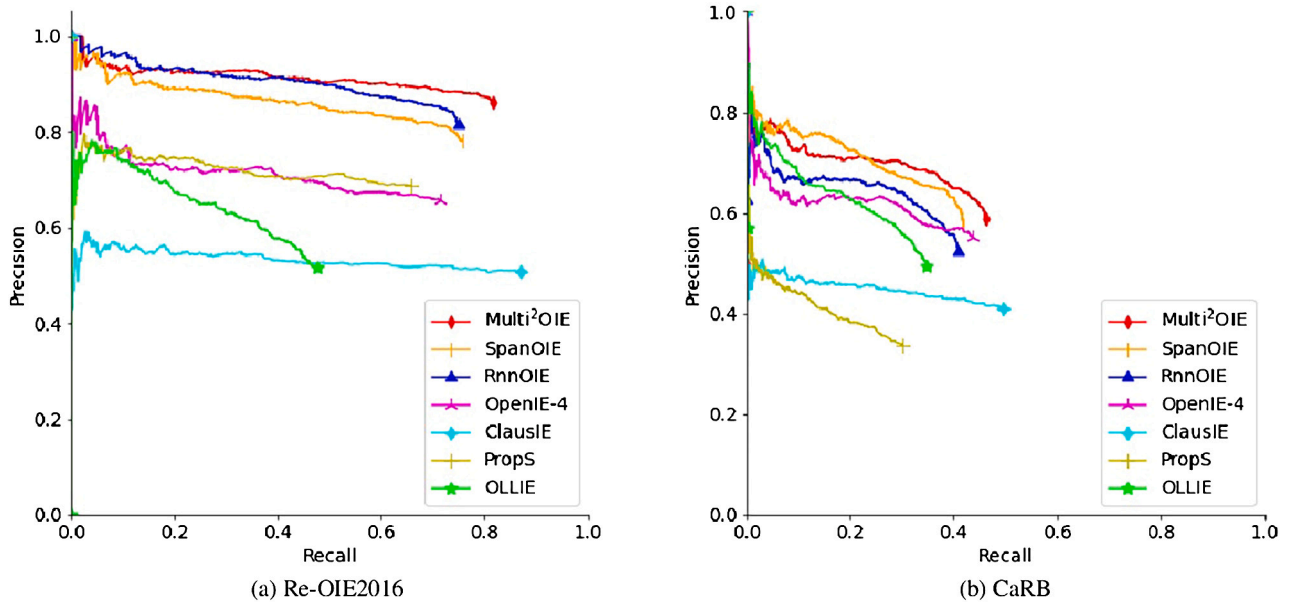
Fig. 14. The architecture of Multi²OIE model (Ro et al. (2020)).Fig. 15. The Performance of Multi²OIE model (Ro et al. (2020)).

Table 8
Relation Extraction dataset Metrics.

Dataset	Domain	#Sentences	#Examples	Binary/n-ary
WEB-500	Web,News	500	461	binary
NYT-500	News	500	150	binary
PENN-100	Mixed	100	51	binary
NYT-222	News,Wiki	222	222	n-ary
OIE2016	Wiki	3200	10359	n-ary
AW-OIE	Wiki, Wikinews	3300	17165	n-ary
ACE2005	Mixed	- ^a	5349	binary
SemEval-2010 Task 8	Mixed	-	10717	binary
LSOIE-wiki	Wiki, Wikinews	24296	56662	n-ary
LSOIE-sci	Science	47998	97550	n-ary
TACRED	Newswire, Web text	- ^b	106,264	binary

^a ACE2005 dataset contains 599 documents containing 5349 relation tuples.

^b TACRED dataset contains approximately same number of sentences as number of examples.

OIE2016 The OIE2016 (Stanovsky and Dagan (2016)) dataset was created by automatic conversion from a question-answering based semantic role labelling database (QA-SRL) (He et al. (2015)). It contains 8477 relation extractions, over 3200 sentences.

AW-OIE The All Words-Open IE (AW-OIE) (Stanovsky et al. (2018)) dataset was created by converting question-answer pairs of the crowd-sourced Question-Answer Meaning Representation (QAMR) dataset. It contains 17,165 extractions spread over 3300 sentences.

CaRB The Crowdsourced automatic Relation extraction Benchmark (CaRB) (Bhardwaj et al. (2019)) dataset was created specifically for OIE by crowd-sourcing, annotating 1282 sentences.

LSOIE The Large Scale Open IE (LSOIE) dataset is converted from the QA-SRL 2.0 dataset in the same manner as OIE2016 (Stanovsky and Dagan (2016)), but it is much larger and more diverse. There are two sections: LSOIE-Wiki from the Wiki/WikiNews domain and LSOIE-Sci from the Science domain. LSOIE contains more than 70,000 sentences and more than 150,000 relation tuples.

TACRED The TAC Relation Extraction Dataset (TACRED) (Zhang, Zhong, et al. (2017)) is one of the largest and most widely used supervised datasets for binary RE. It is built using crowd-sourcing over newswire and web text from the corpus used in the yearly TAC Knowledge Base Population (TAC KBP) challenges between 2009 and 2014. It contains 106,264 examples covering 41 relation types.

5. Domain specific relation extraction

RE system can be applied to wide range of textual sources such as medical papers, legal documents, scientific papers, news etc. With rapidly increasing text documents available, there is an increasing demand to process these documents and store information in the form of structured knowledgebase, which can play important role in downstream applications. RE can help speed up the process of preparing knowledgebase in various domains by automatically extracting relations and providing it to human curators for verification.

Using state-of-the-art models for extracting relations from specific domain text has two main challenges.

- Word representation models such as word2vec Mikolov et al. (2013), GloVe Pennington et al. (2014), ELMo Peters et al. (2018) and BERT Devlin et al. (2019) etc. are trained on text containing general domain text such as Wikipedia. In domain specific corpora word distribution is quite different than the general text, which can degrade the performance of the RE model. To use word embeddings such as word2vec, GloVe, ELMo and BERT need to be trained on particular domain corpora to use in domain specific RE task. Habibi et al. (2017), Lee et al. (2019), Alsentzer, Murphy, Boag, Weng, Jindi, et al. (2019), Chalkidis and Kampas (2019) proposed adapted version of word representations for biomedical, clinical and legal domains.
- Other challenge is state-of-the-art models such as BERT Devlin et al. (2019) is trained on general text, which can not obtain high performance on domain specific text. BERT can achieve good performance on domain specific RE task if we adapt it by training it on particular domain text corpora. Researchers have released BioBERT (Lee et al. (2019)), SciBERT (Beltagy et al. (2019)), ClinicalBERT (Alsentzer, Murphy, Boag, Weng, Jindi, et al. (2019)) and LegalBERT (Chalkidis et al. (2020)) models by fine-tuning BERT model by training on domain specific corpora, which achieved state-of-the-art performance on domain specific RE task. Gu et al. (2021) showed that domain specific training pretraining from scratch substantially outperforms pretraining of general models in the specific domain.

Table 9

Biomedical domain Dataset for RE (Gu et al. (2021)).

Dataset	Train	Dev	Test	Evaluation Metrics
GAD (Bravo et al. (2015))	4261	535	534	Micro F1
EU-ADR (Van Mulligen et al. (2012))	2436	-	-	Micro F1
DDI (Herrero-Zazo et al. (2013))	25296	2496	5716	Micro F1
ChemProt (Krallinger et al. (2017))	18035	11268	15745	Micro F1

EU-ADR dataset contains total 2436 relation examples, which can be used for Train, Dev and Test.

Table 10

Summary of BERT based models in Biomedical Domain (Gu et al. (2021)).

Model	Vocabulary	Pre-training	Corpus	Text Size
BERT	Wiki + Books	-	Wiki + Books	3.3B words / 16 GB
SciBERT	PMC + CS	from scratch	PMC + CS	3.2B words
BioBERT	Wiki + Books	continual pretraining	PubMed	4.5B words
BioBERTv1.1	Wiki + Books	continual pretraining	PubMed + PMC	4.5B words

BioBERTv1.1 model takes BioBERT pre-trained on PubMed + PMC corpus for 10 days as baseline and train it on PubMed for 23 days on Eight NVIDIA V100 (32 GB) GPUs.

5.1. RE in biomedical domain

In this section we discuss Datasets available from Biomedical domain and DNN based models available for Relation Extraction.

5.1.1. Datasets for biomedical domain

Table 9 shows various datasets available for biomedical domain.

- **GAD** (Genetic Association Database) dataset Bravo et al. (2015) contains gene-disease association in Genetic Association Archive.
- **EU-ADR** dataset Van Mulligen et al. (2012) contains annotations on drugs, diseases, genes and proteins and associations between them.
- **DDI** (Drug-Drug Interaction) dataset Herrero-Zazo et al. (2013) contains interaction between Drug-Drug on PubMed abstracts.
- **ChemProt** (Chemical-Protein Interaction) dataset Krallinger et al. (2017) contains PubMed abstracts with annotations of interaction between chemical and protein entities. Dataset contains 23 interactions organized in hierarchy, with 10 high-level interactions (including None).

5.1.2. RE models for biomedical domain

Bhasuran and Natarajan (2018) proposed a model (E-SVM) trained using Ensemble Support Vector Machine for extracting gene-disease relations from biomedical literature. Model used machine learning approach with lexical, syntax, semantic features and word2vec word embeddings trained on biomedical text. Another approach Lim and Kang (2018) uses Recursive Neural Network based model called SPINN (Stack-augmented Parser Interpreter Neural Network). Recent approaches in Biomedical domain uses BERT based model trained on large biomedical literature. Table 10 contains overview of BERT (Devlin et al. (2019)), SciBERT (Beltagy et al. (2019)), BioBERT and BioBERTv1.1 (Lee et al. (2019)). Results of these models are reported in Table 11.

Table 11
Comparison of Biomedical domain RE Performance.

Relation	Dataset		E-SVM	SPINN	BERT	SciBERT	BioBERT	BioBERTv1.1
Gene-Disease	GAD	P	79.21	-	74.28	85.54	76.43	77.32
		R	89.25	-	85.11	80.61	87.65	82.68
		F	83.93	-	79.29	82.83	81.61	79.83
Gene-Disease	EU-ADR	P	76.43	-	75.45	-	80.92	77.86
		R	98.01	-	96.55	-	90.81	83.55
		F	85.34	-	84.62	-	84.83	79.74
Protein-Chemical	ChemProt	P	-	74.80	76.02	79.73	75.20	77.02
		R	-	56.00	71.60	70.85	75.09	75.90
		F	-	64.10	73.74	75.03	75.14	76.46
Disease-Disease	DDI	P	-	-	-	-	-	-
		R	-	-	-	-	-	-
		F	-	-	80.04	81.06	-	80.88

P - Precision, R - Recall and F - F1-Score. The best scores are in bold. Results are referenced from Lee et al. (2019), Hiai et al. (2021) and Gu et al. (2021). Precision and Recall results are not available for DDI dataset.

Table 12
Medical/Clinical domain Dataset for RE.

Dataset	Train	Dev	Test	Evaluation Matrics
i2b2 2010 Uzuner et al. (2011)	3244	-	409	F1-score
n2c2 2018 Henry et al. (2019)	59810	36384	23462	F1-score

5.2. RE in medical domain

In this section we discuss DNN based techniques for Relation Extraction in Medical domain for English language as majority of dataset are available in English. Typical Relations of interest in medial domain are between entities such as Disease, Drug, Frequency, Duration, Dosage, Route, ADE (Adverse Drug Event) and Reason for prescription.

5.2.1. Datasets for medical domain

Table 12 shows different datasets available for Medical domain.

- **i2b2 2010** dataset Uzuner et al. (2011) is a clinical relation extraction dataset proposed in Natural Language Processing Challenges for Clinical Records in 2010. It is annotated by relations between medical problem, treatment and test entities It contains total 3653 relation examples out of which 3244 are for training and 409 are for testing.
- **n2c2 2010** dataset Henry et al. (2019) is a benchmark dataset developed during Track 2 of the 2018 National NLP Clinical Challenges shared task. It is annotated by relations between family members, observations and living status. It focuses on extraction of ADE (Adverse Drug Events) from clinical records.

5.2.2. RE models for medical domain

Table 13 shows the performance of various Deep Learning based models on i2b2 2010 and n2c2 2018 datasets. BERT and BioBERT models already discussed in previous section. ClinicalBERT Alsentzer, Murphy, Boag, Weng, Jin, et al. (2019) is BioBERT model further fine tuned on approx. 2 million notes from Medical Information Mart for Intensive Care-III (MIMIC-III) database Johnson et al. (2016). Wei et al. (2020) uses multi-class classification along with fine tuned BERT model on MIMIC-III database. Mahendran and McInnes (2021) uses binary classification along with fine tuned BERT model. It is evident from the table that Wei et al. (2020), Mahendran and McInnes (2021) produce better results than BERT, BioBERT and ClinicalBERT on n2c2 2018 dataset.

6. Low resource language relation extraction

In resource-rich languages such as English, large amount of labelled corpora is available to train models for different NLP tasks. Low resource languages such as Arabic, Hindi etc. have very limited labelled corpora. Recent advances in Deep learning models for Relation Extractions have achieved promising results; but the model performance drops significantly as the number of training samples decreases. Low resource RE aims to extract relations from text sources where labelled examples are scarce.

One way to solve low resource problem of RE is to use distant supervision Mintz et al. (2009), which assumes that same entity pair appearing in multiple sentences have same relations to augment training data for each relation from external corpus. This approach can be noisy as same entity pair can represent different relation in different context.

Recent approaches use few-shot learning, in which candidate relation statements are ranked and matched in embedding space according to a query relation statement. Few-shot learning problem is defined as, we are given K sets of N labelled relation statements $S = \{(r_0, t_0)...(r_K, t_K)\}$ where $t_i \in 1...N$ is corresponding relation type. Goal is to predict $t_q \in \{1...N\}$ for a given query relation statement r_q . This problem is called N -way K -shot learning Soares et al. (2019). There are various approaches to few-shot RE as follow:

- **Proto** (Han et al. (2018)) is based on Prototyping Networks (Snell et al. (2017)), that tries to find the prototyping vectors for each relation from supporting instances and compares distance between query instance and prototyping vector based on certain distance metrics.
- **BERT-pair** Gao et al. (2019) uses BERT-based model to encode pair of sentences to a probability that the pair of sentences expressing same relation.
- **REGRAB** (Qu et al. (2020)) uses label aware approach that predicts relation label based on similarity between context sentences and the relation label. Relation label presentation is initialized by knowledge graph prepared using Bayesian meta-learning approach. Context sentence representation is learned using BERT_{BASE} model.
- **MTB** Soares et al. (2019) is a pretraining framework assumes that sentences with same head and tail entities are positive pair. During the testing phase, it ranks the similarity score between the query-instance and the support instances and chooses the relation with the highest score as the prediction.
- **CP** Peng et al. (2020) is also a pretraining framework that considers sentences with same relation as positive pairs. It uses same technique as Proto for fine-tuning the model with only difference

Table 13
Comparison of Medical domain RE Performance.

Dataset		BERT	BioBERT	ClinicalBERT	Wei et al. (2020)	Mahendran and McInnes (2021)
i2b2 2010	P	-	-	-	76.24	-
	R	-	-	-	77.34	-
	F	83.50	86.50	86.40	76.79	-
n2c2 2018	P	93.00	93.00	93.00	98.38	93.00
	R	96.00	95.00	90.15	90.00	96.00
	F	94.00	94.00	94.00	94.09	94.00

P - Precision, R - Recall and F - F1-Score.
The best scores are in bold.

Table 14
Comparison results on the test set of the FewRel and NYT-25 datasets in accuracy Dong et al. (2021).

Method	FewRel				NYT-25			
	5-way 1-shot	5-way 5-shot	10-way 1-shot	10-way 5-shot	5-way 1-shot	5-way 5-shot	10-way 1-shot	10-way 5-shot
Proto	80.68	89.60	71.48	82.89	77.63	87.25	66.49	79.51
BERT-pair	88.32	93.22	80.63	87.02	80.78	88.13	72.65	79.68
REGRAB	90.30	94.25	84.09	89.93	89.76	95.66	84.11	92.48
MTB	91.10	95.40	84.30	91.80	88.90	95.53	83.08	92.23
CP	95.10	97.10	91.20	94.70	91.08	94.73	83.99	90.18
MapRE	95.73	97.84	93.18	95.64	91.90	96.01	86.46	92.68

Proto (Han et al. (2018)), BERT-pair (Gao et al. (2019)), REGRAB (Qu et al. (2020)), MTB (Soares et al. (2019)), CP (Peng et al. (2020)).

is it uses dot product instead of Euclidean distance (in the case of Proto) to measure similarity between instances.

- **MapRE** uses similar approach as CP but it uses label-aware approach, in which information from context sentences and their corresponding relation labels are used for pre-training and fine-tuning.

All of these models are evaluated on two few-shot learning benchmarks: FewRel Han et al. (2018) and NYT-25 Gao et al. (2019). The FewRel dataset contains 70,000 sentences for 100 relations (each with 700 sentences) derived from Wikipedia. There are 64 relations for training, 16 relations for validation, and 20 relations for testing. The testing dataset contains 10,000 query sentences that each is given N-way K-shot relation examples and has to be evaluated online (the labels for the testing set are not published). The NYT-25 dataset is a processed dataset for few-shot learning. Qu et al. (2020) suggested a preprocessing strategy to randomly sample 10 relations for training, 5 for validating, and 10 for testing. Table 14 shows the performance of low-resource RE models on FewRel and NYT-25 dataset.

7. Conclusion and future research direction

In this paper, we presented a comprehensive study of Relation Extraction techniques. We first described various traditional approaches to RE. rule-based systems uses a set of rules to extract entities and relations from the text. Most often, these rules are hand-coded, but there are techniques available to automatically learn rules from labelled data. Feature-based methods of supervised RE use a classifier with a set of features to classify Relation instances. In feature-based techniques, the majority of the work goes into defining the right set of features because the efficiency of these features determines how well these systems perform. Kernel-based supervised learning methods use kernel functions to compute similarity between two relations. Kernel-based methods produce better results and require less feature engineering compared to feature-based methods. We also covered the semi-supervised, unsupervised, and distant supervision methods of RE. Recently, DL-based methods are producing state-of-the-art performance for RE task. We also covered the Open IE paradigm and distance supervision-based RE,

which require very little human intervention. There is a continuously increasing trend in RE in the direction of distance supervision based methods. Recent deep learning-based techniques are producing state-of-the-art performances for RE. Currently, CNN-based architectures are widely used for relation extraction and relation classification, but there are very few research papers leveraging RNN architectures for the same.

RE systems have improved a lot in recent years, but they have several problems. We have identified several research directions to improve existing RE systems:

1. English has been the primary language used for RE research, followed by Chinese, Spanish, and Arabic. Most of the datasets released focus on these 4 languages. A detailed study can be carried out about the performance of existing RE methods on other languages, for example on Indian languages with constrained resources (lack of NLP preprocessing tools like POS tagger, parser etc.) and free word order.
2. For extracting n-ary relations, relatively little research has been done. N-ary relations depict relationships between more than two entities. There is a lot of scope for new approaches in this direction.
3. Transfer Learning is a method where models developed for a task are reused for another task. Transfer Learning can provide domain adaptation capability to the Relation Extraction model. BERT is a form of transfer learning that has achieved state-of-the-art performance for many NLP tasks. BERT can be utilised to transfer knowledge from other tasks to the Relation Extraction task.
4. Current research focuses on the extraction of entity pair-relation from a single sentence. In reality, most entities in the text indicate their relationship with one another using many sentences. Some research has proposed methods for cross-sentence RE, but further study is needed to improve the current approaches.

CRedit authorship contribution statement

Kartik Detroja: Formal analysis, Investigation, Resources, Writing – original draft. **C.K. Bhensdadia:** Investigation, Supervision, Writing – review & editing. **Brijesh S. Bhatt:** Investigation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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